

**OKLAHOMA DEPARTMENT OF ENVIRONMENTAL QUALITY
AIR QUALITY DIVISION**

MEMORANDUM

June 5, 2025

TO: Lee Warden, P.E., Permits and Engineering Group Manager

THROUGH: Phillip Martin, P.E., Engineering Manager, Existing Source Permits Section

THROUGH: Junru Wang, P.E., Existing Source Permits Section

FROM: Elijah Ecklund, E.I., Engineering Section, ROAT

SUBJECT: Evaluation of Application No. 2024-1235-C
Plains Marketing, LP
Cushing South Terminal (SIC 4789/NAICS 488999)
Facility ID No: 6854
Latitude: 35.74000°N, Longitude: 96.65120°W
Section 34, Township 15N, Range 6E, Lincoln County, Oklahoma
Directions: From SH-66 and SH-99, go south on SH-99 for approx. 0.75 miles to E Elm St. Turn east, then go approx. 0.70 miles to Richard James Ave. Go north approx. 0.10 miles. Facility is on the west side of the road, the office is on the east side of the road.

SECTION I. INTRODUCTION

Plains Marketing, LP (Plains Marketing or the applicant) has applied for an individual minor source construction permit for their Cushing South Terminal. The facility is currently operating under the Authorization to Operate under the General Permit for Oil and Gas Facilities (GP-OGF) No. 2009-123-O (M-2), issued on October 2, 2024.

Plains Marketing has stated that the purpose of this application is to add additional equipment to the facility. In accordance with Part 3, Section II.C of the GP-OGF, the permittee shall obtain a minor source construction permit for any modification described under Part 1, Section III.C of the GP-OGF. One of the units added is a 60 MMBTUH hot oil heater. This falls under the “new fuel-burning equipment” as defined in OAC 252:100-33 and is applicable to Part 1, Section III.C.10 of the GP-OGF, thus requiring this minor source construction permit.

Based on data provided by Plains Marketing, the facility has emissions of 67.75 TPY of NO_x, 57.45 TPY of CO, 42.98 TPY of VOC, and 1.15 TPY of HAPs, the most significant being 0.31 TPY of H₂CO.

The total emissions from the facility are below the major source thresholds. This facility, therefore, qualifies for a “synthetic minor” permit because the controlled emissions of each of the criteria

pollutants are below the major source threshold of 100 TPY and the HAP emissions are below the 10 TPY threshold for a single HAP and below the 25 TPY threshold for any combination of HAPs.

SECTION II. PROCESS DESCRIPTION

The facility is a crude oil terminal designed to process and store crude oil. Crude oil is offloaded by railcar before entering two (2) 27-MMBTUH boilers, followed by two (2) 16.7-MMBTUH hot oil heaters, and one (1) 60 MMBTUH hot oil heater. Processed crude oil from the heaters is stored in two (2) 70,000-bbl internal floating roof storage tanks before being removed from the facility via pipeline. Three (3) emergency generator engines are on-site to provide backup power as needed.

SECTION III. EQUIPMENT

ID#	Description	Size/Rating	Control Equipment Type	Serial No.	Manufacture, Construction, or Modification Date	Subject to an NSPS or NESHAP Subpart
EMG-ENG1	Caterpillar C-15	762-hp	-	FSE01659	2007	NSPS IIII/ NESHAP ZZZZ
EMG-ENG2	Caterpillar G3512C	1,468-hp	NSCR	TBD	Post-2008	NSPS JJJJ/ NESHAP ZZZZ
EMG-ENG3	Caterpillar G3516C	2,264-hp	NSCR	TBD	Post-2008	NSPS JJJJ/ NESHAP ZZZZ
BOILER1	Boiler	27-MMBTUH	-	-	-	NSPS Dc
BOILER2	Boiler	27-MMBTUH	-	-	-	NSPS Dc
HEAT1	Hot Oil Heater	16.7-MMBTUH	-	-	-	NSPS Dc
HEAT2	Hot Oil Heater	16.7-MMBTUH	-	-	-	NSPS Dc
HEAT3	Hot Oil Heater	60-MMBTUH	-	-	-	NSPS Dc
TANK1	Internal Floating Roof Crude Oil Storage Tank	70,000-bbl	-	-	6/2009	NSPS Kb
TANK2	Internal Floating Roof Crude Oil Storage Tank	70,000-bbl	-	-	6/2009	NSPS Kb
R-LAND	Roof Landings	-	-	-	-	-
FUG	Fugitive Emissions	-	-	-	-	-

SECTION IV. FACILITY-SPECIFIC OR REPRESENTATIVE SAMPLE

TANKS

The flash emissions for the storage tanks were not calculated using a process simulation; therefore, no facility-specific or representative sample is needed for these units.

FUGITIVES

The facility did not submit a gas or liquid sample and assumed 100% VOC content for the gas and liquid service components.

SECTION V. EMISSIONS

Unless otherwise stated, emissions are based on 8,760 hours per year of operation with combustion sources firing field-grade natural gas with a maximum sulfur content of 162 ppmv.

ENGINES

Emissions of NO_x, CO, and VOC from EMG-ENG1 are calculated based on manufacturer data. Emissions of H₂CO are calculated based on AP-42 (4/25) Table 3.4-3 for large uncontrolled stationary diesel engines. H₂CO is included in the VOC emission factors for EMG-ENG1.

Emissions of NO_x, CO, and VOC from EMG-ENG2 and EMG-ENG3 are calculated based on NSPS Subpart JJJJ. H₂CO emissions for EMG-ENG2 and EMG-ENG3 are calculated based on manufacturer data.

Since the NSPS Subpart JJJJ emission limit for VOC does not include H₂CO, H₂CO is added to the VOC emissions shown in the facility-wide emissions summary to represent total VOC for EMG-ENG2 and EMG-ENG3.

EMG-ENG1, EMG-ENG2, and EMG-ENG3 are emergency engines and are calculated based on 500 hours of operation per year.

Engine Emission Factors

ID#	NO _x	CO	VOC	H ₂ CO	
	g/hp-hr	g/hp-hr	g/hp-hr	lb/MMBTU	g/hp-hr
EMG-ENG1 ⁽¹⁾	5.74	0.4	0.01	7.89E-05	-
EMG-ENG2 ⁽²⁾	1.00	2.00	0.70	-	0.15
EMG-ENG3 ⁽³⁾	1.00	2.00	0.70	-	0.15

⁽¹⁾ Fuel consumption is 6,580-BTU/hp-hr

⁽²⁾ Fuel consumption is 6,773-BTU/hp-hr

⁽³⁾ Fuel consumption is 6,632-BTU/hp-hr

Engine Emissions

ID#	NO _x		CO		VOC		H ₂ CO	
	lb/hr	TPY	lb/hr	TPY	lb/hr	TPY	lb/hr	TPY
EMG-ENG1 ⁽¹⁾	9.64	2.41	0.67	0.17	0.02 ⁽²⁾	<0.01 ⁽²⁾	<0.01	<0.01
EMG-ENG2 ⁽¹⁾	3.24	0.81	6.47	1.62	2.76 ⁽²⁾	0.69 ⁽²⁾	0.49	0.12
EMG-ENG3 ⁽¹⁾	4.99	1.25	9.98	2.50	4.24 ⁽²⁾	1.06 ⁽²⁾	0.75	0.19

⁽¹⁾ Emergency Engines emissions are based on 500 hours of operation per year.

⁽²⁾ H₂CO is included in VOC emissions.

HEATERS and BOILERS

Emissions are based on AP-42 (7/98), Section 1.4, a gas heating value of 1,020 BTU/SCF, and the ratings shown in the second table.

Heater/Boiler Emission Factors

ID#	NO _x	CO	VOC
	lb/MMSCF	lb/MMSCF	lb/MMSCF
BOILER1 - BOILER2, HEAT1 - HEAT3	100.0	84.0	5.5

Heater/Boiler Emissions

ID#	Rating	NO _x		CO		VOC	
	MMBTUH	lb/hr	TPY	lb/hr	TPY	lb/hr	TPY
BOILER1	27.0	2.65	11.59	2.22	9.74	0.15	0.64
BOILER2	27.0	2.65	11.59	2.22	9.74	0.15	0.64
HEAT1	16.7	1.64	7.17	1.38	6.02	0.09	0.39
HEAT2	16.7	1.64	7.17	1.38	6.02	0.09	0.39
HEAT3	60.0	5.88	25.76	4.94	21.64	0.32	1.42

TANKS

The crude oil storage tanks are internal floating roof tanks. Working and breathing emissions are composed of rim seal, withdrawal, deck fitting, and deck seam losses. Emissions from the crude oil tanks are calculated using AP-42 (6/20), Section 7.1. VOC emissions from roof landings and cleanings for the tanks were estimated using the TankESP software for non-routine losses. Since ‘weathered’ crude oil is handled for these tanks, no flashing emissions are emitted.

Tank Emissions (per tank)

Parameter	TANK1 – TANK2 Data
Throughput, gal/yr	2,072,616,000
Liquid in Tank(s)	Crude Oil
Working/Breathing Method/Tool	AP-42 (6/20), Section 7.1
Working/Breathing Emissions, TPY	11.00
Landings/Cleaning Emissions, TPY	7.57
Control Type	-
Tank VOC Emitted at Tank, TPY	18.57
Total VOC Emissions, TPY	18.57

FUGITIVES

Emissions from fugitive equipment leaks (FUG) are based on factors obtained from TCEQ’s Air Permit Technical Guidance for Chemical Sources (6/18) Fugitive Guidance, an estimated number of components, and the VOC (C₃+) and HAP content of the materials handled. VOC percent was estimated at 100% for all gases and liquids.

Fugitive Emissions

ID#	VOC, TPY
FUG	0.61

FACILITY-WIDE EMISSIONS**Facility-Wide Emissions**

ID#	Description	NO _x	CO	VOC	HAPs
		TPY	TPY	TPY	TPY
EMG-ENG1	762-hp Caterpillar C-15	2.41	0.17	<0.01 ⁽²⁾	<0.01
EMG-ENG2	1,468-hp Caterpillar G3512C ⁽¹⁾	0.81	1.62	0.69 ⁽²⁾	0.12
EMG-ENG3	2,264-hp Caterpillar G3516C ⁽¹⁾	1.25	2.50	1.06 ⁽²⁾	0.19
BOILER1	27-MMBTUH Boiler	11.59	9.74	0.64	-
BOILER2	27-MMBTUH Boiler	11.59	9.74	0.64	-
HEAT1	16.7-MMBTUH Hot Oil Heater	7.17	6.02	0.39	-
HEAT2	16.7-MMBTUH Hot Oil Heater	7.17	6.02	0.39	-
HEAT3	60-MMBTUH Hot Oil Heater	25.76	21.64	1.42	-
TANK1	70,000-bbl Crude Oil Tank	-	-	18.57	0.42
TANK2	70,000-bbl Crude Oil Tank	-	-	18.57	0.42
FUG	Fugitive Emissions	-	-	0.61	-
Total Emissions		67.75	57.45	42.98	1.15
Emissions from Auth. No. 2009-123-O (M-2)		5.91	32.10	45.64	9.72
Change in Emissions		61.84	25.35	(2.66)	(8.57)

⁽¹⁾ – Equipped with NSCR.

⁽²⁾ – VOC emissions include H₂CO.

HAP EMISSIONS

The internal combustion engines have emissions of hazardous air pollutants, the most significant being formaldehyde. Emissions of formaldehyde for EMG-ENG1 are calculated based on AP-42 (4/25) Table 3.4-3. Emissions of formaldehyde for EMG-ENG2 and EMG-ENG3 are calculated based on manufacture data. Annual emissions are calculated based on 500 hours of operation per year.

Engine Formaldehyde Emissions

ID#	Sources	H ₂ CO	
		lb/hr	TPY
EMG-ENG1	762-hp Caterpillar C-15	<0.01	<0.01
EMG-ENG2	1,468-hp Caterpillar G3512C ⁽¹⁾	0.49	0.12
EMG-ENG3	2,264-hp Caterpillar G3516C ⁽¹⁾	0.75	0.19
Total H₂CO		1.24	0.31

⁽¹⁾ – Equipped with NSCR.

The total HAP emissions from the Plains Marketing equipment at the facility is 1.17 TPY (0.31 TPY of H₂CO). Therefore, the individual and the total emissions of HAPs do not exceed the major source thresholds of 10/25 TPY.

SECTION VI. OKLAHOMA AIR POLLUTION CONTROL RULES

OAC 252:100-1 (General Provisions)

[Applicable]

This Subchapter provides definitions of words and phrases used in Chapter 100.

OAC 252:100-2 (Incorporation by Reference)

[Applicable]

This subchapter incorporates by reference applicable provisions of Title 40 of the Code of Federal Regulations listed in OAC 252:100, Appendix Q. These requirements are addressed in the “Federal Regulations” section.

OAC 252:100-3 (Air Quality Standards and Increments)

[Applicable]

Subchapter 3 enumerates the primary and secondary ambient air quality standards and the significant deterioration increments. At this time, all of Oklahoma is in “attainment” of these standards.

OAC 252:100-5 (Registration, Emissions Inventory and Annual Operating Fees)

[Applicable]

Subchapter 5 requires sources of air contaminants to register with Air Quality, file emission inventories annually, and pay annual operating fees based upon total annual emissions of regulated pollutants. Required annual information (Turn-Around Document) shall be provided to Air Quality.

OAC 252:100-7 (Permits for Minor Facilities)

[Applicable]

Subchapter 7 sets forth the permit application fees and the basic substantive requirements of permits for minor facilities. Since controlled criteria pollutant emissions are less than 100 TPY for each pollutant, and emissions of HAP will not exceed 10 TPY for any one HAP or 25 TPY for any aggregate of HAP, the facility is defined as a “synthetic minor” source.

OAC 252:100-9 (Excess Emissions Reporting Requirements)

[Applicable]

Except as provided in OAC 252:100-9-7(a)(1), the owner or operator of a source of excess emissions shall notify the Director as soon as possible, but no later than 4:30 p.m. the following working day of the first occurrence of excess emissions in each excess emissions event. No later than thirty (30) calendar days after the start of any excess emission event, the owner or operator of an air contaminant source from which excess emissions have occurred shall submit a report for each excess emissions event describing the extent of the event and the actions taken by the owner or operator in response to this event. Request for mitigation, as described in OAC 252:100-9-8, shall be included in the excess emissions event report. Additional reporting may be required in the case of ongoing emission events and in the case of excess emissions reporting required by 40 CFR Parts 60, 61, or 63.

OAC 252:100-13 (Open Burning)

[Applicable]

Open burning of refuse and other combustible material is prohibited except as authorized in the specific examples and under the conditions listed in this subchapter.

OAC 252:100-19 (Particulate Matter)

[Applicable]

Section 19-4 regulates emissions of PM from new and existing fuel-burning equipment, with emission limits based on maximum design heat input rating. Fuel-burning equipment is defined in

OAC 252:100-19 as any internal combustion engine or gas turbine, or other combustion device used to convert the combustion of fuel into usable energy. Thus, the engines, boilers, and heaters are subject to the requirements of this subchapter. OAC 252:100, Appendix C specifies a PM emission limitation of 0.60 lb/MMBTU for all equipment at this facility with a heat input rating of 10 MMBTUH or less. OAC 252:100, Appendix C specifies a PM emission limitation for all equipment at this facility with a heat input rating of greater than 10-MMBTUH, but less than 1,000-MMBTUH based on the following calculation: $E = 1.0428080X^{-0.238561}$, where E is the allowable emission rate and X is the maximum heat input. Table 3.4-1 of AP-42 (4/25) lists total PM emissions from large stationary diesel and all stationary dual-fuel engines to be 0.1 lbs/MMBTU and Table 3.2-3 of AP-42 (7/00) lists the total PM emissions from 4-stroke, rich-burn, natural gas-fired engines to be 0.02 lbs/MMBTU. This permit requires the use of natural gas and ultra-low sulfur diesel for all fuel-burning equipment to ensure compliance with Subchapter 19.

ID#	Equipment	Maximum Heat Input (MMBTUH)	Emissions (lb/MMBTU)	
			Appendix C	Potential
EMG-ENG1	762-hp Caterpillar C-15	5.01	0.60	0.10
EMG-ENG2	1,468-hp Caterpillar G3512C; 4SRB ⁽¹⁾	9.94	0.60	0.02
EMG-ENG3	2,264-hp Caterpillar G3516C; 4SRB ⁽¹⁾	15.01	0.55	0.02
BOILER1	Boiler	27.0	0.48	0.03
BOILER2	Boiler	27.0	0.48	0.03
HEAT1	Hot Oil Heater	16.7	0.53	0.03
HEAT2	Hot Oil Heater	16.7	0.53	0.03
HEAT3	Hot Oil Heater	60.0	0.39	0.03

⁽¹⁾ – Equipped with NSCR.

OAC 252:100-25 (Visible Emissions and Particulates)

[Applicable]

No discharge of greater than 20% opacity is allowed except for short-term occurrences that consist of not more than one six-minute period in any consecutive 60 minutes, not to exceed three such periods in any consecutive 24 hours. In no case shall the average of any six-minute period exceed 60% opacity. When burning natural gas, there is very little possibility of exceeding these standards.

OAC 252:100-29 (Fugitive Dust)

[Applicable]

No person shall cause or permit the discharge of any visible fugitive dust emissions beyond the property line on which the emissions originated in such a manner as to damage or to interfere with the use of adjacent properties, or cause air quality standards to be exceeded, or to interfere with the maintenance of air quality standards. Under normal operating conditions, this facility has negligible potential to violate this requirement; therefore, it is not necessary to require specific precautions to be taken.

OAC 252:100-31 (Sulfur Compounds)

[Applicable]

Part 2 limits the ambient air concentration of H₂S emissions from any facility to 0.2 ppm (24-hour average) at standard conditions which is equivalent to 283 µg/m³. Based on modeling conducted for the general permit for oil and gas facilities, the ambient impacts of H₂S, from oil and gas

facilities combusting natural gas with a maximum H₂S content of 162 ppmv and storing condensate or sweet crude oil, will be in compliance with the ambient air concentration limit.

Part 5 limits sulfur dioxide emissions from new petroleum or natural gas process equipment (constructed after July 1, 1972). For gaseous fuels the limit is 0.2 lb/MMBTU heat input averaged over 3 hours. For fuel gas having a gross calorific value of 1,000 Btu/SCF, this limit corresponds to fuel sulfur content of 1,203 ppmv. Gas produced from oil and gas wells having 162 ppmv or less total sulfur will ensure compliance with Subchapter 31. The permit requires the use of pipeline-grade natural gas or field gas with a maximum sulfur content of 162 ppmv for all fuel-burning equipment to ensure compliance with Subchapter 31.

OAC 252:100-33 (Nitrogen Oxides)

[Not Applicable]

This subchapter limits NO_x emissions from new fuel-burning equipment with rated heat input greater than or equal to 50 MMBTUH. Gas-fired equipment is limited to emissions of 0.20 lbs of NO_x per MMBTU, three-hour average. Compliance with the limit is demonstrated by using continuous emission monitors, stack testing, or AP-42 factors. The one hot oil heater (HEAT3) subject to this subchapter has anticipated NO_x emissions rate of less than or equal to 0.20 lb/MMBTU during normal operations.

OAC 252:100-35 (Carbon Monoxide)

[Not Applicable]

None of the following affected processes are located at this facility: gray iron cupola, blast furnace, basic oxygen furnace, petroleum catalytic cracking unit, or petroleum catalytic reforming unit.

OAC 252:100-37 (Volatile Organic Compounds)

[Applicable]

Part 3 requires storage tanks constructed after December 28, 1974, with a capacity greater than 40,000 gallons to be equipped with either an external floating roof or a fixed roof with an internal-floating cover. The two (2) 2,940,000-gallon crude oil storage tanks are internal floating roof storage tanks and are therefore in compliance.

Part 3 requires loading facilities with a throughput equal to or less than 40,000 gallons per day to be equipped with a system for submerged filling tank trucks or trailers if the capacity of the vehicle is greater than 200 gallons. This facility does not have the physical equipment (loading arm and pump) to conduct this type of loading. Therefore, this requirement is not applicable.

Part 5 limits the organic solvent content of coating of parts and products. Any painting operation will involve maintenance coatings of buildings and equipment and emit less than 100 pounds per day of VOCs and so is exempt.

Part 7 requires fuel-burning and refuse-burning equipment to be operated to minimize emissions of VOC. The equipment at this location is subject to this requirement.

Part 7 requires all effluent water separator openings that receive water containing more than 200 gallons per day of any VOC, to be sealed or the separator to be equipped with an external floating roof or a fixed roof with an internal floating roof or a vapor recovery system. No effluent water separators are located at this facility.

OAC 252:100-42 (Toxic Air Contaminants (TAC))

[Applicable]

This subchapter regulates TAC that are emitted into the ambient air in areas of concern (AOC). Any work practice, material substitution, or control equipment required by the Department prior to June 11, 2004, to control a TAC, shall be retained, unless a modification is approved by the

Director. Since no AOC has been designated there are no specific requirements for this facility at this time.

OAC 252:100-43 (Testing, Monitoring, and Recordkeeping)

[Applicable]

This subchapter provides general requirements for testing, monitoring, and record keeping. It applies to any testing, monitoring, or recordkeeping activity conducted at any stationary source. To determine compliance with emissions limitations or standards, the Air Quality Director may require the owner or operator of any source in the state of Oklahoma to install, maintain, and operate monitoring equipment or to conduct tests, including stack tests, of the air contaminant source. All required testing must be conducted by methods approved by the Air Quality Director and under the direction of qualified personnel. A notice-of-intent to test and a testing protocol shall be submitted to Air Quality at least 30 days prior to any EPA Reference Method stack tests. Emissions and other data required to demonstrate compliance with any federal or state emission limit or standard, or any requirement set forth in a valid permit shall be recorded, maintained, and submitted as required by this subchapter, an applicable rule, or permit requirement. Data from any required testing or monitoring not conducted in accordance with the provisions of this subchapter shall be considered invalid. Nothing shall preclude the use, including the exclusive use, of any credible evidence or information relevant to whether a source would have been in compliance with applicable requirements if the appropriate performance or compliance test or procedure had been performed.

SECTION VII. FEDERAL REGULATIONS

NSPS, 40 CFR Part 60

[Subparts Dc, Kb, IIII, and JJJJ Applicable]

Subpart Dc, Small Industrial-Commercial-Institutional Steam Generating Units. The affected facility to which this subpart applies is each steam generating unit for which construction, modification, or reconstruction is commenced after June 9, 1989, and that has a maximum design heat input capacity between 10 and 100 MMBTUH. The boilers and the hot oil heaters at this facility are within this threshold and are therefore applicable to this subpart.

Subpart Kb, Volatile Organic Liquid Storage Vessels. VOL Storage Vessels. This subpart regulates hydrocarbon storage tanks larger than 19,813 gallons capacity and built after July 23, 1984 and on or before October 4, 2023. The 70,000-bbl Crude Oil Tanks were built in June 2009 and are therefore applicable to this subpart.

Subpart Kc, Volatile Organic Liquid Storage Vessels. VOL Storage Vessels. This subpart regulates hydrocarbon storage tanks larger than 20,000 gallons capacity and built after October 4, 2023. The 70,000-bbl Crude Oil Tanks were built in June 2009 and are therefore not applicable to this subpart.

Subpart GG, Stationary Gas Turbines. There are no compressors at this facility, therefore not applicable.

Subpart VV, Equipment Leaks of VOC in the Synthetic Organic Chemicals Manufacturing Industry (SOCMI). This facility is not a SOCMI plant, therefore not applicable.

Subpart KKK, Equipment Leaks of VOC from Onshore Natural Gas Processing Plans. The facility does not engage in natural gas processing, therefore not applicable.

Subpart LLL, SO₂ Emissions from Onshore Natural Gas Processing. This facility does not have a sweetening unit, therefore not applicable.

Subpart IIII, Stationary Compression Ignition Internal Combustion Engines. The provisions of this subpart are applicable to manufacturers, owners, and operators of stationary compression ignition (CI) internal combustion engines (ICE) constructed after July 11, 2005. EMG-ENG1 is a CI ICE that was manufactured in 2007. Therefore, it is subject to this subpart, and the permit requires compliance with all applicable requirements of this subpart.

The owner/operator of a 2007 model year and later stationary CI-ICE that is certified must comply with the emission standards in Table 2 of Appendix I to Part 1039, Title 40. The applicable standards of Table 2 of Appendix I are listed as follows.

Emission Standards from Table 2, Appendix I, g/kW-hr
Tier 2 Emission Standards

Rated Power (HP)	Starting model year	NMHC + NO_x	CO	PM
560 > kW > 450	2002	6.4	3.5	0.20
kW > 560	2006	6.4	3.5	0.20

Since EMG-ENG1 is a certified engine and manufactured in 2007, the owner/operator must install and configure the engine according to the manufacturer's specifications. Due to being certified, the engine is not required to conduct performance testing. Since EMG-ENG1 is an emergency stationary internal combustion engine, the owner or operator is not required to submit an initial notification. If the emergency engine does not meet the standards applicable to non-emergency engines in the applicable model year, the owner or operator must keep records of the operation of the engine in emergency and non-emergency service that are recorded through the non-resettable hour meter. The owner must record the time of operation of the engine and the reason the engine was in operation during that time. All applicable requirements have been incorporated into the permit.

Subpart JJJJ, Stationary Spark Ignition Internal Combustion Engines. This subpart promulgates emission standards for all new SI engines ordered after June 12, 2006, and all SI engines modified or reconstructed after June 12, 2006, regardless of size. EMG-ENG2 and EMG-ENG3 were manufactured after June 12, 2006. Therefore, they are subject to this subpart, and the permit requires compliance with all applicable requirements of this subpart.

The owner/operator of a stationary SI-ICE with a maximum engine power greater than or equal to 130-HP must comply with the emission standards in Table 1 of Subpart JJJJ. The applicable standards of Table 1 of Subpart JJJJ are listed as follows.

**Emission Standards from Table 1, Subpart JJJJ, g/hp-hr
For Emergency Engines**

Rated Power (HP)	Mfg. Date	NO _x	CO	VOC
≥ 130 HP	-	2.0	4.0	1.0

If EMG-ENG2 and EMG-ENG3 are non-certified engines, the owner/operator must keep a maintenance plan and records of conducted maintenance and must, to the extent practicable, maintain and operate the engine in a manner consistent with good air pollution control practice for minimizing emissions. In addition, the owner/operator must conduct an initial performance test and conduct subsequent performance testing every 8,760 hours of use or 3 years, whichever comes first, thereafter, to demonstrate compliance. Performance tests must comply with the requirements specified in §60.4244. All applicable requirements have been incorporated into the permit.

If EMG-ENG2 and EMG-ENG3 are certified engines, the owner/operator must keep records of conducted maintenance and must, to the extent practicable, maintain and operate the engine in a manner consistent with good air pollution control practice for minimizing emissions, however, no performance testing is required due to the engines being certified. In addition, the owner/operator must keep documentation from the manufacturer that the engine is certified to meet the emission standards and information as required in 40 CFR parts 1048, 1054, and 1060, as applicable. All applicable requirements have been incorporated into the permit.

NESHAP, 40 CFR Part 61

[Not Applicable]

There are no emissions of any of the regulated pollutants: arsenic, asbestos, beryllium, benzene, coke oven emissions, mercury, radionuclides, or vinyl chloride except for trace amounts of benzene. Subpart J (Equipment Leaks of Benzene) concerns only process streams which contain more than 10% benzene by weight. Analysis of Oklahoma natural gas indicates a maximum benzene content of less than 1%.

NESHAP, 40 CFR 63

[Subpart ZZZZ Applicable]

Subpart ZZZZ, Reciprocating Internal Combustion Engines (RICE). This subpart affects RICE that are located at a major source and at area sources. Owners and operators of new or reconstructed engines at area sources and of new or reconstructed engines with a site rating equal to or less than 500 HP located at a major source (except new or reconstructed 4-stroke lean-burn engines with a site rating greater than or equal to 250 HP and less than or equal to 500 HP located at a major source) must meet the requirements of Subpart ZZZZ by complying with either 40 CFR Part 60 Subpart IIII (for CI engines) or 40 CFR Part 60 Subpart JJJJ (for SI engines):

- 1) Stationary RICE located at an area source;
- 2) The following Stationary RICE located at a major source of HAP emissions:
 - i) 2SLB and 4SRB stationary RICE with a site rating of ≤ 500 brake HP;
 - ii) 4SLB stationary RICE with a site rating of < 250 brake HP;
 - iii) Stationary RICE with a site rating of ≤ 500 brake HP which combust landfill or digester gas equivalent to 10% or more of the gross heat input on an annual basis;
 - iv) Emergency or limited use stationary RICE with a site rating of ≤ 500 brake HP; and
 - v) CI stationary RICE with a site rating of ≤ 500 brake HP.

No further requirements apply for engines subject to NSPS under this part. A stationary RICE located at an area source of HAP emissions is new if construction commenced after June 12, 2006. Based on emission calculations, this facility is an area source of HAP. EMG-ENG1 through EMG-ENG3 were constructed after June 12, 2006. They will comply with the requirements of Subpart ZZZZ by complying with NSPS Subpart IIII for EMG-ENG1 and JJJJ for EMG-ENG2 and EMG-ENG3.

Subpart JJJJJJ, National Emission Standards for Hazardous Air Pollutants for Industrial, Commercial, and Institutional Boilers Area Sources. You are subject to this subpart if you own or operate an industrial, commercial, or institutional boiler as defined in § 63.11237 that is located at, or is part of, an area source of HAP, as defined in § 63.2, except as specified in § 63.11195. The facility's boiler is defined as a natural gas-fired boiler. Per § 63.11195 (e), gas-fired boilers are not subject to this part, therefore not applicable.

SECTION VIII. COMPLIANCE

COMPLIANCE AND ENFORCEMENT CASE

There are no active Air Quality compliance or enforcement issues concerning this facility.

INSPECTION

On February 23, 2021, a full compliance evaluation was conducted on-site by DEQ Enforcement Inspector Holly Taber. They were accompanied by Catherine Chinni, Associate Director for Environmental Compliance, who represented Stroud Crude Terminal, LLC (SCT). SCT owned the facility at this time. The ownership of the facility was later transferred to Plains Marketing on May 8, 2024. No violations were discovered.

SECTION IX. ADMINISTRATIVE

TIER CLASSIFICATION AND PUBLIC REVIEW

This application has been determined to be **Tier I** based on the request for a minor source construction permit. The draft permit will undergo public notice on the DEQ's web site as required in OAC 252:4-7-13(g). The public, tribal governments, and the EPA will have 30 days to comment on the draft permit. Permits available for public review and comment are found at this location: <https://www.deq.ok.gov/permits-for-public-review/>.

The applicant has submitted an affidavit that they are not seeking a permit for land use or for any operation upon land owned by others without their knowledge. The affidavit certifies that the applicant owns the land used to accomplish the permitted purpose.

PREVIOUS PERMITS

The facility is currently operating under the GP-OGF Authorization No. 2009-123-O (M-2), issued on October 2, 2024.

FEE PAID

A fee of \$2,000 for an individual construction permit was paid.

SECTION X. SUMMARY

The facility demonstrates the ability to comply with all applicable Air Quality rules and regulations. There are no active Air Quality compliance or enforcement issues that would prevent issuance of the permit. Issuance of the construction permit is recommended, contingent on public review.

**PERMIT TO CONSTRUCT
AIR POLLUTION CONTROL FACILITY
SPECIFIC CONDITIONS**

**Plains Marketing, LP
Cushing South Terminal**

Permit No. 2024-1235-C

The permittee is authorized to construct in conformity with the specifications submitted to Air Quality on December 17, 2024. The Evaluation Memorandum dated June 5, 2025, explains the derivation of applicable permit requirements and estimates of emissions; however, it does not contain operating limitations or permit requirements. Commencing construction under this permit constitutes acceptance of, and consent to, the conditions contained herein:

1. Points of emissions and emissions limitations for each point:

EU ID	Source	NO _x		CO		VOC	
		lb/hr	TPY	lb/hr	TPY	lb/hr	TPY
EMG-ENG1	762 -hp Caterpillar C-15	9.64	2.41	0.67	0.17	0.02 ⁽²⁾	<0.01 ⁽²⁾
EMG-ENG2	1,468-hp Caterpillar G3512C ⁽¹⁾	3.24	0.81	6.47	1.62	2.76 ⁽²⁾	0.69 ⁽²⁾
EMG-ENG3	2,264-hp Caterpillar G3516C ⁽¹⁾	4.99	1.25	9.98	2.50	4.24 ⁽²⁾	1.06 ⁽²⁾
BOILER1	27-MMBTUH Boiler	-	11.59	-	9.74	-	0.64
BOILER2	27-MMBTUH Boiler	-	11.59	-	9.74	-	0.64
HEAT1	16.7-MMBTUH Hot Oil Heater	-	7.17	-	6.02	-	0.39
HEAT1	16.7-MMBTUH Hot Oil Heater	-	7.17	-	6.02	-	0.39
HEAT3	60-MMBTUH Hot Oil Heater	5.88	25.76	4.94	21.64	0.32	1.42
TANK1	70,000-bbl Crude Oil Storage Tank	-	-	-	-	-	37.14
TANK2	70,000-bbl Crude Oil Storage Tank	-	-	-	-	-	

(1) Equipped with NSCR.

(2) Includes H₂CO.

- The facility is authorized to operate continuously (24 hours per day, every day of the year).
- Except for EMG-ENG1, the fuel-burning equipment shall be fired with pipeline grade natural gas or other gaseous fuel with a sulfur content less than 162 ppmv. Engine EMG-ENG1 shall be fueled with diesel fuel with a sulfur content of 0.015% by weight or less. Compliance can be shown by the following methods: for pipeline grade natural gas, a current gas company bill; for other gaseous fuel, a current lab analysis, stain-tube analysis, gas contract, tariff sheet, or other approved methods. Compliance shall be demonstrated at least once each calendar year. For fuels purchased less frequently than once every calendar year, compliance shall be demonstrated for each fuel purchase.
- Each engine at the facility shall have a permanent identification plate attached, which shows the make, model number, and serial number.
- The engines EMG-ENG2 and EMG-ENG3 shall be set to operate with the exhaust gases

passing through a properly functioning NSCR.

6. The emergency engines (EMG-ENG1, EMG-ENG2, and EMG-ENG3) shall be limited to 500 hours of operation per year for each unit and shall be equipped with an hour meter which is either non-resettable or if resettable, equipped with a logbook recording the date and hour meter reading each time the meter is reset. The permittee shall record the number of hours each generator operated each month.
7. The permittee shall conduct an initial test of NO_x and CO emissions from any engine listed in S.C. #1 or any replacement engine; other than (1) an Emergency Use Engine (i.e., any engine that drives an emergency power generator, peaking power generator, firewater pump, or other emergency use equipment and operates no more than 500 hours per year), or (2) any engine equal to or less than 250 horsepower (hp). The initial test must be performed within 180 days of engine startup. Testing shall be conducted using EPA reference methods, if applicable, or a portable analyzer in accordance with a protocol meeting the requirements of the latest AQD "Portable Analyzer Guidance" document, or an equivalent method approved by AQD.
8. At least twice per calendar year, the permittee shall conduct tests of NO_x and CO emissions from any controlled engine greater than 250 hp. Testing shall be conducted using EPA reference methods, if applicable, or a portable analyzer in accordance with a protocol meeting the requirements of the latest AQD "Portable Analyzer Guidance" document, or an equivalent method approved by AQD. Testing is required for any engine that runs for more than 440 hours during a semi-annual period. Each semi-annual test shall be separated by at least 120 days. In the first year of operation, any engine started after March 31st only requires one test regardless of hours operated. The initial test may be counted as the first semi-annual test of an engine.
9. Replacement of any equipment with emissions specified in this permit are authorized under OAC 252:100-7-15(a)(2)(C), provided the replacement unit does not require a change in any emission limit and the owner or operator notifies the DEQ in writing within fifteen (15) days of the startup of the replacement unit. The replacement unit shall meet the definition under OAC 252:100-7-1.1. Installation of an "affected facility," "affected source," or "new source" as those terms are defined in 40 CFR Section 60.2, 40 CFR Section 63.2, and 40 CFR Section 61.02, respectively, that is subject to an emission standard, equipment standard, work practice standard or recordkeeping requirement in a federal NSPS (40 CFR Part 60) or a federal NESHAP (40 CFR Parts 61 and 63) shall comply with all applicable requirements.
10. When periodic compliance testing shows engine exhaust emissions in excess of the lb/hr limits in Specific Condition No. 1, the permittee shall comply with the provisions of OAC 252:100-9 for excess emissions.
11. The crude oil tanks shall be equipped with internal floating roofs. The facility crude oil throughput shall not exceed 4,145,232,000 gallons (based on a 12-month rolling total).

12. For the hot oil heater (HEAT3) listed with a rated heat input of 50 MMBtu/hr or more shall be limited to 0.20 lb/MMBTU, 3-hour rolling average calculated from 3 contiguous hours each hour for any period with 3 hours of complete data, per the requirements outlined in OAC 252:100-33-2(a)(1).
13. The permittee shall comply with all applicable requirements of the NSPS for Small Industrial-Commercial-Institutional Steam Generating Units, Subpart Dc, for each affected steam generating unit, including but not limited to:
 - a. § 60.40c: Applicability and delegation of authority.
 - b. § 60.41c: Definitions.
 - c. § 60.42c: Standard for sulfur dioxide (SO₂).
 - d. § 60.43c: Standard for particulate matter (PM).
 - e. § 60.44c: Compliance and performance test methods and procedures for sulfur dioxide.
 - f. § 60.45c: Compliance and performance test methods and procedures for particulate matter.
 - g. § 60.46c: Emission monitoring for sulfur dioxide.
 - h. § 60.47c: Emission monitoring for particulate matter.
 - i. § 60.48c: Reporting and record keeping requirements.
14. The permittee shall comply with all applicable requirements of the NSPS for Volatile Organic Liquid Storage Vessels, Subpart Kb, for each affected storage vessel, including but not limited to:
 - a. § 60.110b: Applicability and designation of affected facility.
 - b. § 60.111b: Definitions.
 - c. § 60.112b: Standard for volatile organic compounds (VOC).
 - d. § 60.113b: Testing and procedures.
 - e. § 60.114b: Alternative means of emission limitation.
 - f. § 60.115b: Reporting and recordkeeping requirements.
 - g. § 60.116b: Monitoring of operations.
 - h. § 60.117b: Delegation of authority.
15. The permittee shall comply with all applicable requirements of the NSPS for Stationary Compression Ignition Internal Combustion Engines, Subpart IIII, for each affected engine, including but not limited to:
 - a. § 60.4200: Am I subject to this subpart?
 - b. § 60.4201: What emission standards must I meet for non-emergency engines if I am a stationary CI internal combustion engine manufacturer?
 - c. § 60.4202: What emission standards must I meet for emergency engines if I am a stationary CI internal combustion engine manufacturer?
 - d. § 60.4203: How long must my engines meet the emission standards if I am a manufacturer of stationary CI internal combustion engines?
 - e. § 60.4204: What emission standards must I meet for non-emergency engines if I am an owner or operator of a stationary CI internal combustion engine?

- f. § 60.4205: What emission standards must I meet for emergency engines if I am an owner or operator of a stationary CI internal combustion engine?
 - g. § 60.4206: How long must I meet the emission standards if I am an owner or operator of a stationary CI internal combustion engine?
 - h. § 60.4207: What fuel requirements must I meet if I am an owner or operator of a stationary CI internal combustion engine subject to this subpart?
 - i. § 60.4208: What is the deadline for importing or installing stationary CI ICE produced in previous model years?
 - j. § 60.4209: What are the monitoring requirements if I am an owner or operator of a stationary CI internal combustion engine?
 - k. § 60.4210: What are my compliance requirements if I am a stationary CI internal combustion engine manufacturer?
 - l. § 60.4211: What are my compliance requirements if I am an owner or operator of a stationary CI internal combustion engine?
 - m. § 60.4212: What test methods and other procedures must I use if I am an owner or operator of a stationary CI internal combustion engine with a displacement of less than 30 liters per cylinder?
 - n. § 60.4213: What test methods and other procedures must I use if I am an owner or operator of a stationary CI internal combustion engine with a displacement of greater than or equal to 30 liters per cylinder?
 - o. § 60.4214: What are my notification, reporting, and recordkeeping requirements if I am an owner or operator of a stationary CI internal combustion engine?
 - p. § 60.4215: What requirements must I meet for engines used in Guam, American Samoa, or the Commonwealth of the Northern Mariana Islands?
 - q. § 60.4216: What requirements must I meet for engines used in Alaska?
 - r. § 60.4217: What emission standards must I meet if I am an owner or operator of a stationary internal combustion engine using special fuels?
 - s. § 60.4218: What General Provisions and confidential information provisions apply to me?
 - t. § 60.4219: What definitions apply to this subpart?
16. The permittee shall comply with all applicable requirements of the NSPS for Stationary Spark Ignition Internal Combustion Engines, Subpart JJJJ, for each affected engine, including but not limited to the following:
- a. §60.4230: Am I subject to this subpart?
 - b. §60.4231: What emission standards must I meet if I am a manufacturer of stationary SI internal combustion engines?
 - c. §60.4232: How long must my engines meet the emissions standards if I am a manufacturer of stationary SI internal combustion engines?
 - d. §60.4233: What emission standards must I meet if I am an owner or operator of a stationary SI internal combustion engine?
 - e. §60.4234: How long must I meet the emissions standards if I am an owner or operator of a stationary SI internal combustion engine?
 - f. §60.4235: What fuel requirements must I meet if I am an owner or operator of a stationary SI internal combustion engine?

- g. §60.4236: What is the deadline for importing or installing stationary SI ICE produced in the previous model year?
 - h. §60.4237: What are the monitoring requirements if I am an owner or operator of a stationary SI internal combustion engine?
 - i. §60.4238: What are my compliance requirements if I am a manufacturer of stationary SI internal combustion engines < 19 KW (25 HP).
 - j. §60.4239: What are my compliance requirements if I am a manufacturer of stationary SI internal combustion engines > 19 KW (25 HP) that use gasoline?
 - k. §60.4240: What are my compliance requirements if I am a manufacturer of stationary SI internal combustion engines > 19 KW (25 HP) that use LPG?
 - l. §60.4241: What are my compliance requirements if I am a manufacturer of stationary SI internal combustion engines participating in the voluntary certification program?
 - m. §60.4242: What other requirement must I meet if I am a manufacturer of stationary SI internal combustion engines?
 - n. §60.4243: What are my compliance requirements if I am an owner or operator of a stationary SI internal combustion engine?
 - o. §60.4244: What test methods and other procedures must I use if I am an owner or operator of a stationary SI internal combustion engine?
 - p. §60.4245: What are my notification, reporting, and recordkeeping requirements if I am an owner or operator of a stationary SI internal combustion engine?
 - q. §60.4246: What parts of the General Provisions apply to me?
 - r. §60.4247: What parts of the mobile source provisions apply to me if I am a manufacturer of stationary SI internal combustion engines?
 - s. §60.4248: What definitions apply to this subpart?
17. The permittee shall comply with all applicable requirements of the NESHAP for Stationary Reciprocating Internal Combustion Engines (RICE), Subpart ZZZZ, for each affected engine, including but not limited to:
- a. § 63.6580: What is the purpose of subpart ZZZZ?
 - b. § 63.6585: Am I subject to this subpart?
 - c. § 63.6590: What parts of my plant does this subpart cover?
 - d. § 63.6595: When do I have to comply with this subpart?
 - e. § 63.6603: What emission limitations, operating limitations, and other requirements must I meet if I own or operate an existing stationary RICE located at an area source of HAP emissions?
 - f. § 63.6605: What are my general requirements for complying with this subpart?
 - g. § 63.6612: By what date must I conduct the initial performance tests or other initial compliance demonstrations?
 - h. § 63.6615: When must I conduct subsequent performance tests?
 - i. § 63.6620: What performance tests and other procedures must I use?
 - j. § 63.6625: What are my monitoring, installation, operation, and maintenance requirements?
 - k. § 63.6630: How do I demonstrate initial compliance with the emission limitations, operating limitations, and other requirements?
 - l. § 63.6635: How do I monitor and collect data to demonstrate continuous compliance?

- m. § 63.6640: How do I demonstrate continuous compliance with the emission limitations, operating limitations, and other requirements?
 - n. § 63.6645: What notifications must I submit and when?
 - o. § 63.6650: What reports must I submit and when?
 - p. § 63.6655: What records must I keep?
 - q. § 63.6660: In what form and how long must I keep my records?
 - r. § 63.6665: What parts of the General Provisions apply to me?
 - s. § 63.6670: Who implements and enforces this subpart?
 - t. § 63.6675: What definitions apply to this subpart?
18. The permittee shall maintain records of operations as listed below. These records shall be maintained on-site or at a local field office for at least five years after the date of recording and shall be provided to regulatory personnel upon request.
- a. Periodic testing for NO_x and CO emissions from the engines if required.
 - b. Operating hours for the emergency engines (monthly and 12-month rolling total).
 - c. For the fuels burned, maintain the appropriate document(s) as described in Specific Condition No. 3.
 - d. Records required under NESHAP 40 CFR Part 60, Subparts Dc, Kb, IIII, and JJJJ.
 - e. Records required under NESHAP 40 CFR Part 63, Subpart ZZZZ.
 - f. Facility crude oil throughput (monthly and 12-month rolling total).
19. The permittee shall submit an application for an operating permit within 180 days if commencement of operation of any emission source whose construction has been authorized by this permit.

**MINOR SOURCE PERMIT TO OPERATE / CONSTRUCT
AIR POLLUTION CONTROL FACILITY
STANDARD CONDITIONS
(February 13, 2020)**

- A. The issuing Authority for the permit is the Air Quality Division (AQD) of the Oklahoma Department of Environmental Quality (DEQ) in accordance with and under the authority of the Oklahoma Clean Air Act. The permit does not relieve the holder of the obligation to comply with other applicable federal, state, or local statutes, regulations, rules, or ordinances. This specifically includes compliance with the rules of the other Divisions of DEQ: Land Protection Division and Water Quality Division.
- B. A duly issued construction permit or authorization to construct or modify will terminate and become null and void (unless extended as provided in OAC 252:100-7-15(g)) if the construction is not commenced within 18 months after the date the permit or authorization was issued, or if work is suspended for more than 18 months after it is commenced. [OAC 252:100-7-15(f)]
- C. The recipient of a construction permit shall apply for a permit to operate (or modified operating permit) within 180 days following the first day of operation. [OAC 252:100-7-18(a)]
- D. Unless specified otherwise, the term of an operating permit shall be unlimited.
- E. Notification to the Air Quality Division of DEQ of the sale or transfer of ownership of this facility is required and shall be made in writing by the transferor within 30 days after such date. A new permit is not required. [OAC 252:100-7-2(f)]
- F. The following limitations apply to the facility unless covered in the Specific Conditions:
1. No person shall cause or permit the discharge of emissions such that National Ambient Air Quality Standards (NAAQS) are exceeded on land outside the permitted facility. [OAC 252:100-3]
 2. All facilities that emit air contaminants are required to file an emission inventory and pay annual operating fees based on the inventory. Instructions are available on the Air Quality section of the DEQ web page. www.deq.ok.gov [OAC 252:100-5]
 3. Deviations that result in emissions exceeding those allowed in this permit shall be reported consistent with the requirements of OAC 252:100-9, Excess Emission Reporting Requirements. [OAC 252:100-9]
 4. Open burning of refuse and other combustible material is prohibited except as authorized in the specific examples and under the conditions listed in the Open Burning subchapter. [OAC 252:100-13]
 5. No particulate emissions from new fuel-burning equipment with a rated heat input of 10 MMBTUH or less shall exceed 0.6 lbs/MMBTU. [OAC 252:100-19]
 6. No discharge of greater than 20% opacity is allowed except for short-term occurrences which consist of not more than one six-minute period in any consecutive 60 minutes, not to exceed three such periods in any consecutive 24 hours. In no case shall the average of any six-minute period exceed 60% opacity. [OAC 252:100-25]

7. No visible fugitive dust emissions shall be discharged beyond the property line on which the emissions originate in such a manner as to damage or to interfere with the use of adjacent properties, or cause air quality standards to be exceeded, or interfere with the maintenance of air quality standards. [OAC 252:100-29]
8. No sulfur oxide emissions from new gas-fired fuel-burning equipment shall exceed 0.2 lbs/MMBTU. No existing source shall exceed the listed ambient air standards for sulfur dioxide. [OAC 252:100-31]
9. Volatile Organic Compound (VOC) storage tanks built after December 28, 1974, and with a capacity of 400 gallons or more storing a liquid with a vapor pressure of 1.5 psia or greater under actual conditions shall be equipped with a permanent submerged fill pipe or with an organic material vapor-recovery system. [OAC 252:100-37-15(b)]
10. All fuel-burning equipment shall at all times be properly operated and maintained in a manner that will minimize emissions of VOCs. [OAC 252:100-37-36]

G. Any owner or operator subject to provisions of NSPS shall provide written notification as follows: [40 CFR 60.7 (a)]

1. A notification of the date construction (or reconstruction as defined under §60.15) of an affected facility is commenced postmarked no later than 30 days after such date. This requirement shall not apply in the case of mass-produced facilities which are purchased in completed form.
2. A notification of any physical or operational change to an existing facility which may increase the emission rate of any air pollutant to which a standard applies, unless that change is specifically exempted under an applicable subpart or in §60.14(e). This notice shall be postmarked 60 days or as soon as practicable before the change is commenced and shall include information describing the precise nature of the change, present and proposed emission control systems, productive capacity of the facility before and after the change, and the expected completion date of the change. The Administrator may request additional relevant information subsequent to this notice.
3. A notification of the actual date of initial start-up of an affected facility postmarked within 15 days after such date.
4. If a continuous emission monitoring system is included in the construction, a notification of the date upon which the test demonstrating the system performance will commence, along with a pretest plan, postmarked no less than 30 days prior to such a date.

H. Any owner or operator subject to provisions of NSPS shall maintain records of the occurrence and duration of any start-up, shutdown, or malfunction in the operation of an affected facility or any malfunction of the air pollution control equipment. [40 CFR 60.7 (b)]

I. Any owner or operator subject to the provisions of NSPS shall maintain a file of all measurements and other information required by this subpart recorded in a permanent file suitable for inspection. This file shall be retained for at least five years following the date of such measurements, maintenance, and records. [40 CFR 60.7 (f)]

J. Any owner or operator subject to the provisions of NSPS shall conduct performance test(s) and furnish to AQD a written report of the results of such test(s). Test(s) shall be conducted within 60 days after achieving the maximum production rate at which the facility will be operated, but not later than 180 days after initial start-up. [40 CFR 60.8]



PERMIT

AIR QUALITY DIVISION
STATE OF OKLAHOMA
DEPARTMENT OF ENVIRONMENTAL QUALITY
707 N. ROBINSON, SUITE 4100
P.O. BOX 1677
OKLAHOMA CITY, OKLAHOMA 73101-1677

Permit No. 2024-1235-C

Plains Marketing, LP

having complied with the requirements of the law, is hereby granted permission to construct the Cushing South Terminal, located Section 34, Township 15N, Range 6E, Lincoln County, Oklahoma, subject to standard conditions dated February 13, 2020, and specific conditions, both attached.

In the absence of construction commencement, this permit shall expire 18 months from the issuance date, except as authorized under Section B of the Standard Conditions.

DRAFT

Lee Warden, P.E.
Permits and Engineering Group Manager

Date Issued

Richard S. Anderson
Plains Marketing, LP
333 Clay Street, Suite 1600
Houston, TX 77002

SUBJECT: Minor Source Construction Permit No. **2024-1235-C**
Plains Marketing, LP
Cushing South Terminal
Facility ID: 6854
Section 34, Township 15N, Range 6E, Lincoln County, Oklahoma

Dear Richard S. Anderson:

Enclosed is the permit authorizing the construction of the facility referenced above. Please note that this permit is issued subject to standard and specific conditions, which are attached. These conditions must be carefully followed since they define the limits of the permit and will be confirmed by periodic inspections.

Also note that you are required to annually submit an emissions inventory for this facility. An emissions inventory must be completed through DEQ's electronic reporting system by April 1st of every year. Any questions concerning the submittal process should be referred to the Emissions Inventory Staff at (405) 702-4100.

Thank you for your cooperation. If you have any questions, please refer to the permit number above and contact me at Elijah.Ecklund@deq.ok.gov or by phone at (918) 293-1622.

Sincerely,

Elijah Ecklund

Elijah Ecklund
Engineering Section
AIR QUALITY DIVISION

Enclosures: AQD Acronym

Department of Environmental Quality (DEQ)
Air Quality Division (AQD)
Acronym List
11-21-2024

ACFM	Actual Cubic Feet per Minute	GAL	Gallon (gal)
AD	Applicability Determination	GDF	Gasoline Dispensing Facility
AFRC	Air-to-Fuel Ratio Controller	GEP	Good Engineering Practice
API	American Petroleum Institute	GHG	Greenhouse Gases
ASTM	American Society for Testing and Materials	GR	Grain(s) (gr)
AVO	Audio, Visual, or Olfactory	H₂CO	Formaldehyde
BACT	Best Available Control Technology	H₂S	Hydrogen Sulfide
BAE	Baseline Actual Emissions	HAP	Hazardous Air Pollutants
BBL	Barrel(s)	HC	Hydrocarbon
BHP	Brake Horsepower (bhp)	HCFC	Hydrochlorofluorocarbon
BTEX	Benzene, Toluene, Ethylbenzene, Xylene	HFR	Horizontal Fixed Roof
BTU	British thermal unit (Btu)	HON	Hazardous Organic NESHAP
		HP	Horsepower (hp)
		HR	Hour (hr)
C&E	Compliance and Enforcement	I&M	Inspection and Maintenance
CAA	Clean Air Act	IBR	Incorporation by Reference
CAM	Compliance Assurance Monitoring	ICE	Internal Combustion Engine
CAS	Chemical Abstract Service		
CAAA	Clean Air Act Amendments	LAER	Lowest Achievable Emission Rate
CC	Catalytic Converter	LB	Pound(s) [Mass] (lb, lbs, lbm)
CCR	Continuous Catalyst Regeneration	LB/HR	Pound(s) per Hour (lb/hr)
CD	Consent Decree	LDAR	Leak Detection and Repair
CEM	Continuous Emission Monitor	LNG	Liquefied Natural Gas
CFC	Chlorofluorocarbon	LT	Long Ton(s) (metric)
CFR	Code of Federal Regulations	LPE	Legally and Practicably Enforceable
CI	Compression Ignition		
CNG	Compressed Natural Gas	M	Thousand (Roman Numeral)
CO	Carbon Monoxide or Consent Order	MAAC	Maximum Acceptable Ambient Concentration
COA	Capable of Accommodating	MACT	Maximum Achievable Control Technology
COM	Continuous Opacity Monitor	MM	Prefix used for Million (Thousand-Thousand)
D	Day	MMBTU	Million British Thermal Units (MMBtu)
DEF	Diesel Exhaust Fluid	MMBTUH	Million British Thermal Units per Hour (MMBtu/hr)
DG	Demand Growth	MMSCF	Million Standard Cubic Feet (MMscf)
DSCF	Dry Standard (At Standard Conditions) Cubic Foot (Feet)	MMSCFD	Million Standard Cubic Feet per Day
EGU	Electric Generating Unit	MSDS	Material Safety Data Sheet
EI	Emissions Inventory	MWC	Municipal Waste Combustor
EPA	Environmental Protection Agency	MWe	Megawatt Electrical
ESP	Electrostatic Precipitator		
EUG	Emissions Unit Group	NA	Nonattainment
EUSGU	Electric Utility Steam Generating Unit	NAAQS	National Ambient Air Quality Standards
FCE	Full Compliance Evaluation	NAICS	North American Industry Classification System
FCCU	Fluid Catalytic Cracking Unit	NESHAP	National Emission Standards for Hazardous Air Pollutants
FIP	Federal Implementation Plan		
FR	Federal Register	NH₃	Ammonia
GACT	Generally Achievable Control Technology	NMHC	Non-methane Hydrocarbon

NGL	Natural Gas Liquids	SCFM	Standard Cubic Feet per Minute
NO₂	Nitrogen Dioxide	SCR	Selective Catalytic Reduction
NO_x	Nitrogen Oxides	SER	Significant Emission Rate
NOI	Notice of Intent	SI	Spark Ignition
NSCR	Non-Selective Catalytic Reduction	SIC	Standard Industrial Classification
NSPS	New Source Performance Standards	SIP	State Implementation Plan
NSR	New Source Review	SNCR	Selective Non-Catalytic Reduction
O₃	Ozone	SO₂	Sulfur Dioxide
O&G	Oil and Gas	SO_x	Sulfur Oxides
O&M	Operation and Maintenance	SOP	Standard Operating Procedure
O&NG	Oil and Natural Gas	SRU	Sulfur Recovery Unit
OAC	Oklahoma Administrative Code	T	Tons
OC	Oxidation Catalyst	TAC	Toxic Air Contaminant
OGI	Optical Gas Imaging	TEG	Triethylene Glycol
PAH	Polycyclic Aromatic Hydrocarbons	THC	Total Hydrocarbons
PAE	Projected Actual Emissions	TPY	Tons per Year
PAL	Plant-wide Applicability Limit	TRS	Total Reduced Sulfur
Pb	Lead	TSP	Total Suspended Particulates
PBR	Permit by Rule	TV	Title V of the Federal Clean Air Act
PCB	Polychlorinated Biphenyls	µg/m³	Micrograms per Cubic Meter
PCE	Partial Compliance Evaluation	US EPA	U. S. Environmental Protection Agency
PEA	Portable Emissions Analyzer	VFR	Vertical Fixed Roof
PFAS	Per- and Polyfluoroalkyl Substance	VMT	Vehicle Miles Traveled
PM	Particulate Matter	VOC	Volatile Organic Compound
PM_{2.5}	Particulate Matter with an Aerodynamic Diameter <= 2.5 Micrometers	VOL	Volatile Organic Liquid
PM₁₀	Particulate Matter with an Aerodynamic Diameter <= 10 Micrometers	VRT	Vapor Recovery Tower
POM	Particulate Organic Matter or Polycyclic Organic Matter	VRU	Vapor Recovery Unit
ppb	Parts per Billion	YR	Year
ppm	Parts per Million	2SLB	2-Stroke Lean Burn
ppmv	Parts per Million Volume	4SLB	4-Stroke Lean Burn
ppmvd	Parts per Million Dry Volume	4SRB	4-Stroke Rich Burn
PSD	Prevention of Significant Deterioration		
psi	Pounds per Square Inch		
psia	Pounds per Square Inch Absolute		
psig	Pounds per Square Inch Gage		
RACT	Reasonably Available Control Technology		
RATA	Relative Accuracy Test Audit		
RAP	Regulated Air Pollutant or Reclaimed Asphalt Pavement		
RFG	Refinery Fuel Gas		
RICE	Reciprocating Internal Combustion Engine		
RO	Responsible Official		
ROAT	Regional Office at Tulsa		
RVP	Reid Vapor Pressure		
SCC	Source Classification Code		
SCF	Standard Cubic Foot		
SCFD	Standard Cubic Feet per Day		